CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A (80 MARKS)

QUESTION 1

a) Explain two characteristics of Trojan horse malware, including how it disguises itself and what harm it can cause once active on a host.

(2 marks)

b) Explain six techniques by which malware propagates itself or attaches its code to a host program, and describe the effect each technique has on the infected system.

(6 marks)

QUESTION 2

a) State TWO (2) differences between morals and laws as forms of conduct guidance for computing professionals.

(2 marks)

b) Describe six (6) professional conduct obligations that the ACM code places on a computing professional, and explain why such a documented code is needed to guide conduct in the computing profession.

(6 marks)

QUESTION 3

a) Explain the two stages of the biometric authentication process and state what happens during each stage.

(2 marks)

b) State six facts about biometric mechanisms and technologies as presented in the course, covering the definition of a biometric, the two application sectors, the behavioural examples, the physiological examples, the fingerprint process, and the iris scanning advantages.

(6 marks)

QUESTION 4

a) State TWO (2) classes of threat and, for each, the type of harm it causes to a system's assets.

(2 marks)

b) The Samy worm spread through MySpace by exploiting a cross-site scripting flaw. Explain how this worm injected its payload, how it harvested and recruited each victim's friends, and the scale of the resulting infection.

(6 marks)

QUESTION 5

a) Explain two ethical impacts that technology can have on society, and describe how a professional code of ethics helps computing professionals respond to such impacts.

(2 marks)

b) Refer to your answer in (a). Based on the scenario in (a), discuss how the computing professional's responsibility for privacy should be exercised, referring to the ACM code of ethics and professional conduct and to the responsibilities of information specialists working within ethical, moral, and legal constraints.

(6 marks)

QUESTION 6

a) Explain the Chinese Wall model as a formal security model, stating the concern it is designed to address and how its access decisions differ from static policies.

(2 marks)

b) Compare the access control matrix, access control lists and capability lists as structures for implementing access policies, and explain the requirements that any such access control structure must satisfy.

(6 marks)

QUESTION 7

a) Explain the goals that access control pursues and the role of a reference monitor in realizing access control policies.

(2 marks)

b) State six points about the formal security models covered in the lecture, covering the BLP and Biba models, the Chinese Wall model, and the classification of the Chinese Wall policy.

(6 marks)

QUESTION 8

a) State TWO (2) facts about the chain of trust in trusted computing, including how measurements are recorded and where the chain begins.

(2 marks)

b) Explain the roles of the two primary system components of the Next Generation Secure Computing Base (NGSCB), describing what each component does and how they relate to the trusted platform.

(6 marks)

QUESTION 9

a) List TWO (2) key points about software insecurity and buffer overflows as described in the Software Security lecture.

(2 marks)

b) Discuss six (6) aspects of Windows security mechanisms, covering the purpose of its access control facilities, the role of Active Directory and tokens, the historical security milestones, and the risks that arise from the complexity of these mechanisms.

(6 marks)

QUESTION 10

a) List two ways in which an intrusion detection system can help defend a system against attacks, according to the lecture.

(2 marks)

b) Describe three classes of threat to a computer system and, for each class, state one countermeasure that the lecture associates with preventing, detecting or recovering from it.

(6 marks)

PART B (20 MARKS)

QUESTION 1

A system administrator is reviewing how a shared UNIX server protects its user accounts. She wants to strengthen the controls applied to the stored password files and to the way accounts behave after repeated failed logins, while keeping the login process usable for staff.

a) Discuss FIVE (5) password security controls that can be applied to protect user accounts on a shared system, as presented in the course.

(5 marks)

b) The administrator in the scenario is considering single sign-on for the shared UNIX server. Explain how Kerberos and related single sign-on mechanisms address authentication in such an environment, giving FIVE (5) points as presented in the course.

(5 marks)

QUESTION 2

a) Explain the architecture of a Network-Based Intrusion Detection System by describing how sensors should be placed and what strengths this architecture offers the campus network team.

(5 marks)

b) State five preventive network security mechanisms that a campus network team could deploy, giving for each the specific protective role it plays in controlling or securing network traffic.

(5 marks)

END OF QUESTION PAPER